MENG30011 Applied Solid Mechanics

Summative Coursework - FEA

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1 Single Element Finite Analysis

1.1 Element Geometry and Definition

1.1.1 Defining the Element

A single 2D continuum element was used to model the lug, according to the brief, whose type was a 6-node (second order) triangular element, resulting in 12 degrees of freedom, as each node was free to displace in the x and y directions. The resulting displacement vector for a 6-node element with 12 DoF is as shown:

$$\mathbf{u} = \left[u_{1x}, u_{1y}, u_{2x}, u_{2y}, u_{3x}, u_{3y}, u_{4x}, u_{4y}, u_{5x}, u_{5y}, u_{6x}, u_{6y}\right]^{\mathrm{T}} \tag{1}$$

The lug was heavily bending dominated as it was clamped at one end and loaded at the other, similar to a cantilever beam. Quadratic elements are capable of capturing bending stresses through a linearly varying strain field (useful for capturing the stress gradient along the weld line), whereas a constant strain triangle (CST) assumes a constant strain field across the element, justifying the selection of a second order element. The limitation of using a second order element is that more algebra is required than for a CST (necessary in this case). The single element initial approximation cannot capture localised stresses of high concentrations (around the lug eye and weld line); however, for an initial approximation of the lug's global bending stresses and displacements, the single element model using a second order element to capture bending stresses was appropriate.

1.1.2 Node Placement

Figure 1 depicts the chosen node locations for the triangular element overlaid on the lug and a representation of the force and boundary conditions. The selected nodes cover the entire weld line (the location where high stresses are expected, excluding the lug eye) and set a third node at the location of the applied point force through the lug. In this model, the weld line was idealised as perfectly rigid, meaning nodes 1, 3 and 6 have 0 displacement in the x and y directions, an assumption that may underestimate the true flexion of the weld by approximating a stiffer model. Node 2 is the location of the applied point force, and nodes 4 and 5 are free to displace in both x and y. This element gives a simple, initial approximation for the lug response but will likely underestimate the displacements due to the weld line nodes being fixed, when in reality they would have some flexibility.

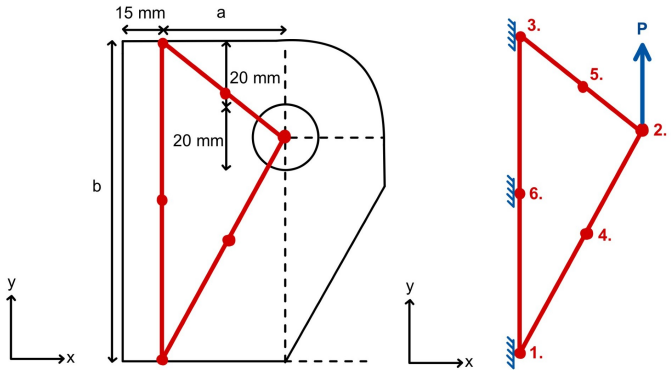


Figure 1: (a) Second Order Element and Boundary Conditions.

1.1.3 Modelling Assumptions Summary

- A point force is applied through the centre of the lug eye of the model, as if a perfect pin acts through the lug eye.
- Boundary conditions at nodes 1, 3 and 6 are perfectly rigid, so the boundary conditions had 0 displacement (this will result in an underestimate for displacement).

- 2D plane stress as the lug geometry in thickness (c) is relatively thin and the lug is loaded in plane.
- Linear elastic, isotropic S355JR structural steel.
- Small strains and small displacements (expected to remain elastic and, therefore, deflections must be small for steel).

1.2 Shape Functions

1.2.1 Defining Geometry and Shape Functions

Using the a, b and c values extracted from the provided python script (Appendix A.1), the nodal co-ordinates for the element are as shown.

Node 1: $(x_1, y_1) = (0, 0)$,
Node 2: $(x_2, y_2) = (27, 69)$,
Node 3: $(x_3, y_3) = (0, 84)$,
Node 4: $(x_4, y_4) = (13.5, 34.5)$,
Node 5: $(x_5, y_5) = (13.5, 76.5)$,
Node 6: $(x_6, y_6) = (0, 42)$.

Figure 2 shows how the element and nodes were isoparametrically mapped from the physical space (x,y) to the natural space (s,t). This was done so the shape functions could interpolate the displacement and geometry so the irregular triangular element could be represented in natural co-ordinates.

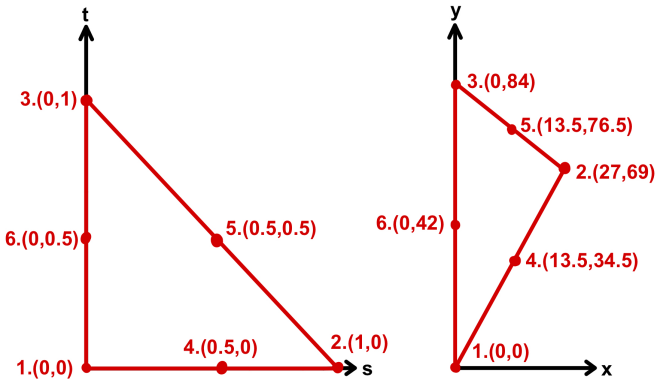


Figure 2: Isoparametric Mapping.

$$\mathbf{N}(s,t) = \begin{bmatrix} N_1 & 0 & N_2 & 0 & \dots & N_6 & 0 \\ 0 & N_1 & 0 & N_2 & \dots & 0 & N_6 \end{bmatrix} \tag{2}$$

Using the standard 6-node triangle shape functions from the notes [2] (Appendix A.2), the N matrix was constructed, equation (2), where the first row interpolates u_x and the second u_y. This N matrix was used to calculate the displacement interpolation vector:

$$\begin{bmatrix} u_x \\ u_y \end{bmatrix} = \mathbf{N}(s,t)\,\mathbf{u} \tag{3}$$

1.2.2 Jacobian Matrix

To form the expressions for strain, we know:

$$\begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{bmatrix} = \begin{bmatrix} \dfrac{\partial}{\partial x} & 0 \\ 0 & \dfrac{\partial}{\partial y} \\ \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial x} \end{bmatrix} \mathbf{N}\,\mathbf{u} = \mathbf{B}\,\mathbf{u}. \tag{4}$$

So, the strain-displacement matrix $\mathbf{B}(s,t)$ [2] (Appendix A.3) (evaluated at the 3 Gauss quadrature points in section 1.2.4) was formed in terms of x and y derivatives. However, this could not be solved before transforming the differentials from physical (x,y) co-ordinates into natural (s,t) co-ordinates using the Jacobian and inverse Jacobian matrices (Appendix A.4). These Jacobian matrices were used throughout to transform co-ordinate systems. They could be evaluated using the expressions for $x(s,t)$ and $y(s,t)$ (Appendix A.5). The strain values still could not be solved until the nodal displacement vector had known values. To ensure the $\mathbf{J}$ matrix was non-degenerate, the $\det(\mathbf{J})$ was computed and found to be positive at all Gauss points.

1.2.3 Plane Stress D matrix

The triangular element was also modelled in plane stress as the lug was relatively thin through the thickness ($c < b$), and the dominant loading is applied in plane which allowed for the out of plane stress components to be assumed to be negligible. This meant the 3x3 2D D matrix for plane stress was used [2].

$$\mathbf{D}^{(2D)} = \frac{E}{1-\nu^2}\begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \dfrac{1-\nu}{2} \end{bmatrix} \tag{5}$$

Substituting the values for E and ν from the material properties table (Appendix A.6) gave a numerical matrix (Appendix A.7) that could be used as a constant when evaluating the integral. While appropriate for capturing global bending responses, out-of-plane bending stresses and displacements are not captured.

1.2.4 Stiffness Matrix Evaluation

The element stiffness matrix could be calculated using the equation shown below.

$$\mathbf{K} = \iiint_V \mathbf{B}^{\mathrm{T}}(\mathbf{q})\,\mathbf{D}\,\mathbf{B}(\mathbf{q})\,\mathrm{d}V(\mathbf{r}) \tag{6}$$

Equation 6 first needed to be evaluated in natural co-ordinates to simplify the integration. The determinant of the J matrix (Appendix A.4) ($|J|$) was used to transform the integral. Applying Gauss quadrature theory for a triangular element using the standard Gauss points [2] for a second order triangular element of $(0.5, 0)$, $(0, 0.5)$, $(0.5, 0.5)$, and the weightings of $1/6$ for each point, the integral was broken down into the form:

$$\mathbf{K} = \frac{1}{6}\,t\,|J_1|\mathbf{B}_1^{\mathrm{T}}\mathbf{D}\mathbf{B}_1 + \frac{1}{6}\,t\,|J_2|\mathbf{B}_2^{\mathrm{T}}\mathbf{D}\mathbf{B}_2 + \frac{1}{6}\,t\,|J_3|\mathbf{B}_3^{\mathrm{T}}\mathbf{D}\mathbf{B}_3 \tag{7}$$

Where $|J_i|$ is the determinant of the Jacobian matrix at the Gauss points (Appendix A.4) used to translate to natural co-ordinates as previously discussed, t is the thickness (c for this scenario), and $\mathbf{B}$ is the calculated matrix (Appendix A.3) in terms of s and t at each of the Gauss points. For the displayed 12×12 stiffness matrix (Appendix A.8), the numerical values for the material properties were left as algebraic so a dimensionless $\mathbf{K}$ matrix could be evaluated to check it was correct. The matrix was checked to be symmetric (as expected for a linear elastic element) and have 9 positive eigenvalues (as a result of the elastic deformational modes) and 3 Eigenvalues that were approximately 0 (representing the rigid-body translation and rotation) which were expected for the 12 DoFs and 3 rigid body modes.

1.3 Boundary Conditions and Applied Load

1.3.1 Applied Load

From the information provided in the brief, the pressure vessel mass is 12,000 kg which is distributed across 6 identical lugs. The design also requires a safety factor of 3 which must be accounted for at this stage of calculation to provide the worst allowable case of loading. First the weight of the pressure vessel was calculated using equation (8).

$$W = mg = 12{,}000 \times 9.81 = 117{,}720 \text{ N} \tag{8}$$

Which split over six lugs is:

$$F_{\text{lug}} = \frac{W}{6} = 19{,}620 \text{ N} \tag{9}$$

And applying the safety factor of 3 from the brief gave the design load per lug:

$$F_{\text{safe}} = 3F_{\text{lug}} = 58{,}860 \text{ N} \tag{10}$$

After applying these equations, the maximum load per lug was calculated as 58,860 N, in this case, applied vertically at node 2 (as shown in Figure 1).

1.3.2 Boundary Conditions

Referring to Figure 1, the boundary conditions were modelled along the weld line meaning that nodes 1, 3 and 6 have 0 displacement. So, the global displacement vector was reduced to:

$$\mathbf{u} = \left[0, 0, u_{2x}, u_{2y}, 0, 0, u_{4x}, u_{4y}, u_{5x}, u_{5y}, 0, 0\right]^{\mathrm{T}} \tag{11}$$

Node 2 is subject to the full force calculated above, so loading vertically upward at node 2:

$$\mathbf{F} = \left[0, 0, 0, P, 0, 0, 0, 0, 0, 0, 0, 0\right]^{\mathrm{T}} \tag{12}$$

where $P = 58{,}860$ N.

1.4 Nodal Displacements

With the fixed DoFs for displacement, the stiffness matrix was reduced for the remaining free DoFs and material property values (Appendix A.6) were substituted into the $\mathbf{K}$ matrix. This gave the reduced stiffness matrix (Appendix A.9) and when substituted into equation (13):

$$\mathbf{F} = \mathbf{K}\,\mathbf{u} \tag{13}$$

Gave expressions for the nodal deformations (Appendix A.10). Reading from the deformation vector shows the maximum vertical displacement (displacement at node 2 at (27,69) mm) to be calculated as 1.11×10^{-5} m or 0.0111 mm, upward due to positive sign convention.

1.5 Stress and Strain

With nodal displacements having been calculated, the Gauss quadrature points could be substituted into the expression for strain to give the 3 expressions for plane and shear strains using equation (4). This gave the 3 sets of strains at the 3 Gauss quadrature points (Appendix A.11). Substituting the numerical 3×3 plane-stress $\mathbf{D}$ matrix (Appendix A.7) and the corresponding strain matrices at the Gauss points (Appendix A.11) into equation (14),

$$\boldsymbol{\sigma} = \mathbf{D}\,\boldsymbol{\varepsilon} \tag{14}$$

The stress matrix for each of these Gauss points was calculated, and using equation (15), the von Mises stress at these Gauss points was also calculated (Appendix A.12).

$$\sigma_{\mathrm{vm}} = \sqrt{\sigma_{xx}^2 - \sigma_{xx}\sigma_{yy} + \sigma_{yy}^2 + 3\tau_{xy}^2} \tag{15}$$

1.6 Interpreting results

1.6.1 Nodal Displacement

As calculated before, the second node moves 1.11×10^{-5} m upwards which is small compared to the lengths of a, b and c, verifying the small strain and displacement assumption made previously. To sanity check the displacement matrix, a strain energy consistency check was done comparing the strain energy and work done equations (Appendix A.13). Both were found to be equal to 0.327 J, meaning $U = W$, and the nodal displacements were assumed to be accurate for this model. When compared to a cantilever Euler-Bernoulli beam, tip deflection was found using this equation:

$$u_{\max} = \frac{-PL^3}{3EI} \tag{16}$$

Where $P = 58{,}860$ N, L was approximated as length $(a = 27$ mm$)$ and E was found in the material properties table (Appendix A.6). When approximating the base as $(c = 34$ mm$)$ and height as $(b = 84$ mm$)$ to find I for this case (Appendix A.14), the vertical displacement for this approximation was calculated as $u_{\max} = 0.02$ mm, greater by a factor of 2 which is extremely close considering the crude approximation, and even closer considering the underestimated value for deflection the FE model predicts.

1.6.2 Strain Values

The values at the quadrature points (Appendix A.11), calculated by substituting Gauss points and displacement information into equation (4). Gauss point 1 is dominated by the positive ε_{xx} tensile values and Gauss point 3 is mostly dominated by the negative ε_{yy} compressive value representing the lower surface in tension and the upper surface in compression which intuitively makes sense for this scenario. As Gauss point 2 was set as a fixed boundary condition, the strain values for this point are all equal to 0. Whilst this may not be accurate for the real lug, these 0 values were caused because of the perfectly rigid weld assumption.

1.6.3 Stress Values

There was maximum stress at Gauss point 1, (13.5,34.5) mm in physical co-ordinates, near the loaded node 2, with a von Mises stress of 97 MPa (Appendix A.12), however, due to this initial approximation only being a single element model, the local peak stress values (around the lug eye) were averaged resulting in an underestimated stress reading. Compared to the maximum allowable yield stress for this material of 355 MPa (Appendix A.6), the lug design, at least according to the initial approximation of the triangular element, is only deformed elastically, well short of the value for plastic deformation, and is well within the safe loading conditions at a safety factor of 3. Again, due to the quadrature point 2 having 0 strain, due to fixed displacement at the weld boundary, the calculated stress for this model was 0, an unrealistic value in reality with non-idealised boundary conditions. To ensure safe loading conditions are met, a multi-element FEA model should be done to check in higher resolution the stress values within the lug.

2 Abaqus FEA

Abaqus FEA software [3] was used to analyse the lug in more detail using a multi-element model. This analysis aimed to produce more accurate values for deformation and stresses within the lug to see locations of high stress and discern whether the lug can truly be considered safe under the load it is subject to.

2.1 Geometry and Loading

2.1.1 Part Geometry

Using the geometry from Figure 1 and the a, b and c values (Appendix A.1) the part for the lug and the base plate were modelled on Abaqus. Considering manufacturing details of the lug, a 5 mm chamfer was designed at the

welded corner to meet BS EN 1090-2 [4] standards and to reduce local stress concentrations near the weld. The same elastic material properties were applied to the weld geometry as the lug and base to represent a compatible filler metal. The thermal histories were assumed to be the same as the rest of the lug and the weld was modelled as perfect (no residual stresses from welding). This assumption may result in an underestimate for peak stress.

2.1.2 Boundary and Loading Conditions

The rear face of the back plate was fully constrained (U1 = U2 = U3 = UR1 = UR2 = UR3 = 0), modelling the base plate to be attached rigidly to the pressure vessel. This assumption may lead to high stress values at the element layer in contact with the boundary condition, which may misrepresent a high concentration of stress at a point, when in reality the stress would be distributed into the pressure vessel; something that should be accounted for when discussing results. A vertical traction force was applied through the top partitioned face of the lug eye to the lug to approximate a lifting pin force in contact with the upper surface of the lug eye (without having to design a pin component), however a force applied in this way may cause stress peaks in the lug eye due to contact shear. The magnitude of this force was 58,860 N as calculated in Section 1.3.1, which included the safety factor of 3 from the brief.

2.1.3 Modelling Assumptions Summary

- Linear elastic, isotropic S355JR structural steel (according to the brief and material properties provided, Appendix A.6). Plasticity occurs after yield stress of 355 MPa (where the lug is considered to fail).
- Force acts on the upper surface of the lug eye and is idealised as a traction force. When using an idealised force, the contact region inside the lug eye may produce a singularity (stresses tending to infinity as the mesh is refined).
- Perfect weld between the lug and the back plate, so thermal and material properties at the weld are identical to the rest of the lug. This neglects residual stresses from welding and any deformities in the weld in reality.
- Fully fixed back plate, neglecting stress distribution at boundary into the pressure vessel potentially resulting in stress singularities at the boundary.
- Load is solely vertical and from the loading cables (no self weight) so stresses at different loading vectors are not considered. This means further failure analysis on the lug will have to be performed to ensure safety at different loading angles.

These assumptions were made to allow simplifications during modelling that improve computational efficiency whilst still capturing the bending dominated lug behaviour. The material assumptions were discussed in more detail in the section below.

2.2 Material Properties

The material properties table (Appendix A.6) was used to assign density (ρ = 8010 kg m^{-3}), elastic properties of Young's Modulus (E = 209 GPa) and Poisson's ratio (ν = 0.3), and plastic properties acting from the yield stress (σ_y = 355 MPa) to the ultimate tensile strength (σ_{uts} = 630 MPa) after 22% elongation. Which meant physically, any stress up to 355 MPa would result in elastic non-permanent deformation and any value above this would result in permanent plastic deformation, and for this design be unacceptable.

2.3 Mesh Convergence Study

To reduce computational processing and time, a specific element type was selected to test in seed size study. This eliminated the need to run multiple convergence studies through logical selection and elimination of element types, saving on time and processing power. During the seed study, the values were considered to converge when the percentage change between seed sizes reduced to less than 3%.

2.3.1 Mesh Selection

Whilst hex elements are effective and more accurate with respect to degrees of freedom, for the lug's complex geometry consisting of curved profiles, the lug eye, and triangular prismatic welds, hex (and wedge) elements would struggle to effectively mesh around these areas. Whilst weld partitions and other geometries could be made to accommodate these element types, this would misrepresent the lug and would result in distorted elements around the welds and curved profiles. Tetrahedral elements are ideal for these complex geometries and were therefore selected. First order tetrahedral elements, as previously discussed in section 1, are overly stiff and would not capture the bending dominated forces through the lug; a C3D10 (second order, quadratic) tetrahedral mesh was used for the convergence study and selected as the most appropriate element type. One limitation with quadratic tetrahedral elements is their high computational processing required at small seed sizes, something to consider during mesh refinement.

2.3.2 Seed Size Study

Addressing the concern for computational processing, a global seed size of 10 mm was assigned to the lug, and for each seed size tested, a local edge refinement was assigned to each of the points of interest along the weld lines. The inner surface of the lug eye was not considered as a point of interest, as the stresses were misrepresented due to the artificial idealised contact force, as the brief suggests. Therefore, the surface of the lug eye was removed and stress was analysed for the remaining elements as displayed visually in Figure 3.

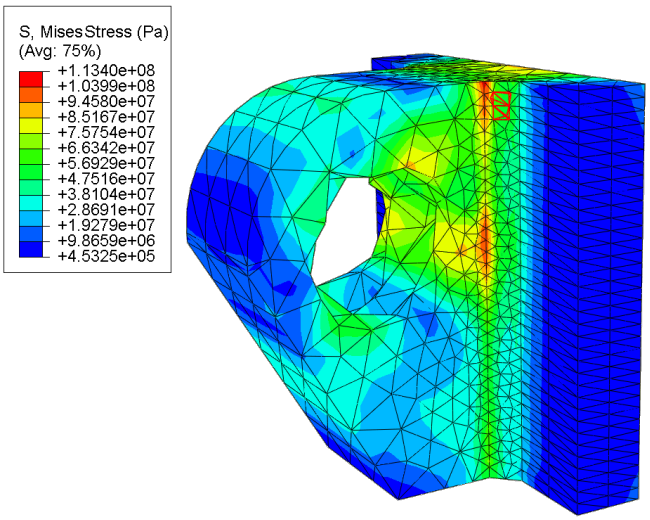
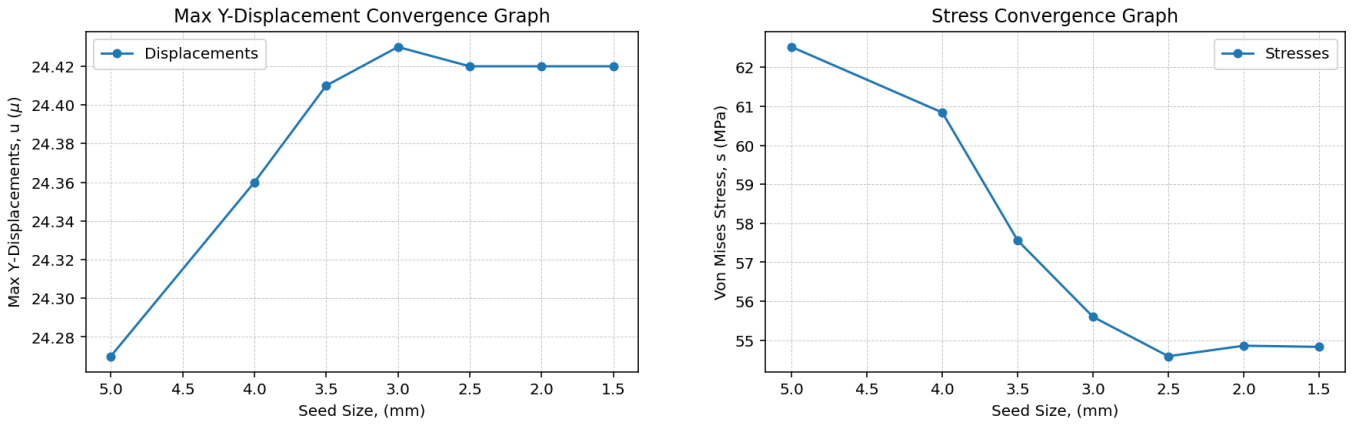


Figure 3: Patch Average and Mesh Visualisation

Maximum displacement on the top of the lug eye converged simply as the mesh was refined; however, the von Mises stresses were harder to analyse. At two points on the weld line there were singularities, whereas the mesh was refined further the stress at the singularity node tended to infinity due to re-entrance corners on the weld line (the maximum values for stress would diverge). To get a von Mises stress convergence, a patch average stress of a known area away from the singularity on the top of the weld line was taken, as highlighted in Figure 3. A partitioned-face of constant area (to avoid results being affected by the changing seed size) was created in the part module, and the average nodal stress across the set was taken.
Using the values for maximum displacement and patch average stress from the set in Figure 3, Figures 4a and 4b were plotted to show convergence from seed sizes of 5 to 1.5 mm (for the edges of interest).
The seed size at which both graphs converged (variations ¡3%) was 3 mm. The seed size selected for optimal analysis was 2.5 mm where the patch stress changed by 1.8%; at this point, convergence had settled and the mesh was refined enough to produce an accurate approximation for stresses, without requiring extensive computational processing.



(a) Displacement Convergence

(b) von Mises Stress Convergence

Figure 4: Convergence Graphs

2.4 Assessing Results

A C3D10 quadratic tetrahedral mesh with a seed size of 2.5 mm was chosen for subsequent analysis. When mesh quality was checked, Abaqus showed no highly distorted elements of Jacobian ratios less than 0.4. Overall, the deformed shape with compression at the top and tension at the bottom of the lug indicated a global bending response from the lug around the weld line.

2.4.1 Vertical Deformation

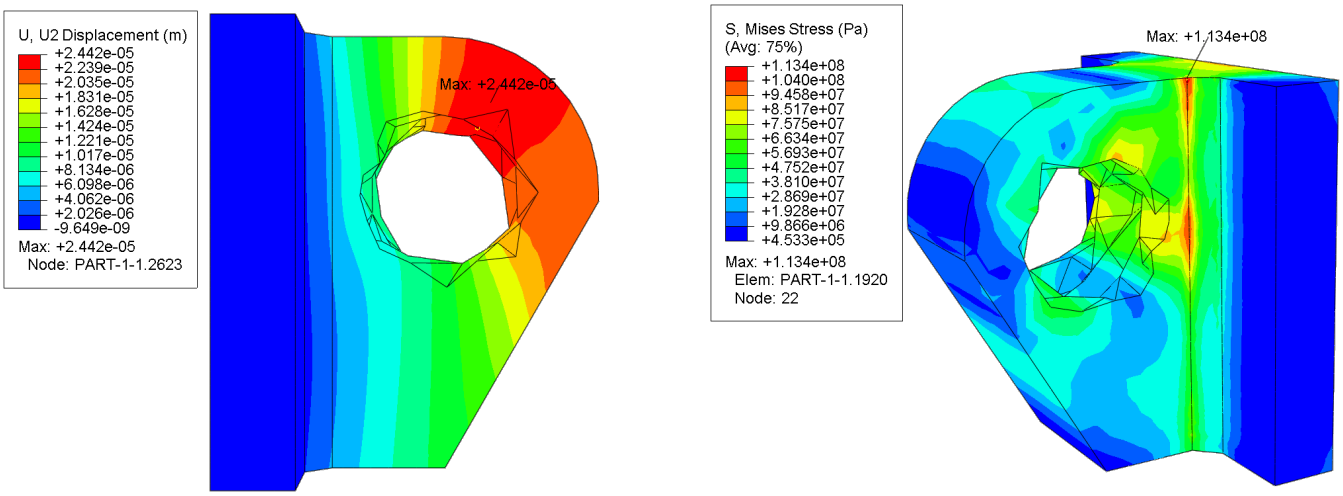
The maximum vertical nodal displacement (excluding the lug eye's inner surface) was 24.42 μm at node 2623 above the lug eye within the upper section of the lug as shown in Figure 5a. This value makes sense as the node closest to the lug surface is subject to the most force and furthest from fixed supports and, therefore, would have maximum deformation (a similar principle to the cantilever beam analogy used in section 1).

2.4.2 Von Mises Stress

The von Mises stress showed revealed several regions of high stress on the model. First, when the inner surface of the eye was excluded, a maximum von Mises stress of 113.4 MPa occurred at the top of the weld line in the singularity previously discussed at node 22 (Figure 5b) where huge compressive stresses occur due to the lug's bending behaviour. While the true value is uncertain due to this node being a singularity, this overestimation can be used in comparison to safety criteria as a conservative estimate. The centre of the weld line is under high shear stress and the top, rear of the base plate showed a small peak in compressive stress at the last element-layer (Figure 5b). This is due to the fully constrained boundary layer assigned to the rear, meaning all of the compressive stress is concentrated at this rear layer, when in reality the stress would be dissipated into the pressure vessel that would have some flexibility. This could be resolved using an elastic boundary or designing the pressure vessel, however, for this analysis the lug is the focus.

2.4.3 Design Adequacy

The calculated peak von Mises stress of 113.4 MPa (though an overestimation from a singularity node) sits well below the yield stress of 355 MPa (32%) even with the load accounting for a safety factor of three, suggesting the lug will remain in a purely elastic state with no plasticity or fracture expected under the design load. Potential fracture requires crack like defects, analysed in section 3. The calculated maximum von Mises stress occurred at the weld line as expected from section 1 for a bending dominated lug. Comparing analytical results, the calculated maximum von Mises stress was 97 MPa and the maximum deformation was 11 μm and compared to the FEA results (113.4



(a) Maximum Vertical Deformation

(b) Maximum von Mises Stress

Figure 5: Maximum Deformation and Von Mises Stress

MPa and 24.42 μm) show values of the same factor with slight underestimates (14% and 45% respectively). This was expected (as discussed in section 1) as the single element model models an overly stiff weld with no back plate, and could not capture the 3D detailed bending. The multi-element model in section 2 provided a much more reliable result. Based on the loading criteria provided, the lug design is safe, but further failure analysis of crack formation and potential misuse scenarios should be performed.

3 Failure Analysis

This section assessed further safety criteria for the lug and commented on its performance and overall safety. As the lug surface was being analysed in this section, a plane-stress assumption could be made.

3.1 Plastic Deformation

This section assessed the magnitude and direction of an additional stress σ_p from oscillations and cable misalignment, as stated in the brief, that would cause yield to be reached and permanent plastic deformation to occur. The location of highest stress on the lug (and therefore the first point to reach yield stress), as seen in Figure 5b, was around node 22 where the lug connects to the back plate; this node could not be used for analysis, however, due to it being a singularity point so a nearby high stress node was selected. For this analysis, node 14,802 shown in Figure 6 (Appendix B.11) was selected.

$$\sigma_0 = \begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{bmatrix} = \begin{bmatrix} -77.58 & -24.53 \\ -24.53 & -7.215 \end{bmatrix} MPa \tag{17}$$

The reduced plane-stress matrix (17) for node 14,802 was formed using values extracted from Abaqus using the plane-stress assumption previously stated. This matrix was superimposed with the matrix (Appendix B.1), derived from the plane-stress transformation equations [1], to give the combined stress matrix for an additional load σ_p at angle θ_p.

$$\sigma_{\text{tot}} = \begin{bmatrix} \sigma_{11} + \sigma_p \cos^2\theta_p & \sigma_{12} + \sigma_p \sin\theta_p \cos\theta_p \\ \sigma_{21} + \sigma_p \sin\theta_p \cos\theta_p & \sigma_{22} + \sigma_p \sin^2\theta_p \end{bmatrix}. \tag{18}$$

Using a python script, expressions in terms of σ_p and θ_p were substituted into the expressions for principal stresses (Appendix B.2), then into the equation for von Mises stresses in plane-stress (Appendix B.5), and set equal to the yield stress (σ_y = 355 MPa). Finally, a sinusoidal graph of θ_p against σ_p was plotted (Appendix B.12). From this

graph, the minimum σ_p (269.47 MPa) at angle (θ_p = 17.5°) was obtained, giving the magnitude of additional stress aligned with the principal stress resulting in permanent plastic deformation at the critical location.

3.2 Strain Gauge

The strain gauge was attached at node 14,802, where compressive von Mises stress was a maximum due to the global bending response. Using a compressive strain gauge and assuming a linear stress-strain relationship, a critical strain could be measured and used to warn of yielding. Physically, the area that the strain gauge was applied needed to fit a 3-6 mm strain gauge [5] and be accessible for attachment. To find the most sensitive orientation of the strain gauge, the operational and critical case from the previous section were assessed.

3.2.1 Strain at Operational Loading Conditions

The strain values (from Abaqus) on node 14,802 (Figure 6, Appendix B.11) under operational load were:

$$\varepsilon_{xx} = -357.4\mu\varepsilon, \quad \varepsilon_{yy} = 80.33\mu\varepsilon, \quad \gamma_{xy} = -305.1\mu\varepsilon$$

Principal strains and the angle at which they act were obtained using standard plane-stress relations [1, 2] (Appendix B.3,B.4).

$$\varepsilon_{1op} = 128\mu\varepsilon, \quad \varepsilon_{2op} = -405\mu\varepsilon, \quad \theta_{op} = 17.5^\circ$$

Meaning the maximum absolute measured strain during normal loading conditions, at orientation θ_{op} = 17.5°, was ε_{2op} = -405$\mu\varepsilon$.

3.2.2 Critical Strain at Yield

To calculate the measured critical strain at the yield point, the values for σ_p and θ_p from section 3.1 (269.47 MPa and 17.5° respectively) were substituted into matrix (18) to give the critical stresses σ_x, σ_y and τ_{xy} shown below.

$$\sigma_x = -322.7MPa, \quad \sigma_y = -31.6MPa, \quad \tau_{xy} = -101.8MPa$$

The general stress-strain relations (Appendix B.7) were used to calculate the strains under critical loading.

$$\varepsilon_x = -1491\mu\varepsilon, \quad \varepsilon_y = 311\mu\varepsilon, \quad \gamma_{xy} = -1260\mu\varepsilon,$$

When applied to the principal strain equations (Appendix B.3) gave a value for the measured critical compressive strain ε_{2crit} = -1688$\mu\varepsilon$ at an angle θ_{crit} = 17.5°. Which intuitively makes sense, as the angles of applied additional stress and strain and the operational stress should be equal to maximise the superimposed stress.

3.2.3 Strain Gauge Orientation

To find the most sensitive orientation of strain gauge, the differences of ε_x, ε_y, and γ_{xy} were found below.

$$\Delta\varepsilon_x = \varepsilon_{x,\text{crit}} - \varepsilon_{x,\text{op}} = -1491\mu\varepsilon + 357.4\mu\varepsilon = -1134\mu\varepsilon, \tag{19}$$

$$\Delta\varepsilon_y = \varepsilon_{y,\text{crit}} - \varepsilon_{y,\text{op}} = 311\mu\varepsilon - 80\mu\varepsilon = 231\mu\varepsilon, \tag{20}$$

$$\Delta\gamma_{\text{xy}} = \gamma_{\text{xy,crit}} - \gamma_{\text{xy,op}} = -1260\mu\varepsilon + 305\mu\varepsilon = -955\mu\varepsilon. \tag{21}$$

Taking the difference between operational and critical strains and applying the principal strain angle formula shows the most sensitive orientation is θ_{crit} = 17.5°. Aligning a strain gauge at this angle will give a reading of ε_2 = -1688 $\mu\varepsilon$.

3.3 Brittle Fracture

This section considers cracks forming under Mode I loading, meaning crack forming due to stress acting perpendicular to crack propagation. It was also assumed, from the brief, that the normal to the crack surface is in the x-y plane and the magnitude of additional stresses that cause the crack were limited by the critical stress σ_p found in section 3.1. Following these criteria, the crack would form at the location on the lug under the highest tension, which was at the bottom portion of the lug surface where the weld corner is.

3.3.1 Orientation of σ_p

From Abaqus, the stress values at the selected node 14,495 for analysis were:

$$\sigma_x = 66.1MPa, \quad \sigma_y = 11.3MPa, \quad \tau_{xy} = -15.2MPa$$

Using these values and equation 49 (Appendix B.4) the angle of principal stress was found to be θ = -13°. In theory, the most dangerous orientation of an additional stress σ_p would be when it is superimposed at the same orientation as the stress experienced by the operational load (as shown in section 3.2). For this reason, θ_p can be assumed to be oriented at an angle of -14.5° (the same as the operational stress). Assuming this load is limited to the magnitude calculated in section 3.1 of (269.47 MPa), as specified in the brief, the magnitude of the stress on this node is equal to the sum of the principal stresses experienced from the operational and additional loads. This normal stress that would cause brittle fracture and a mode I crack is $\sigma_n = 339.52$ MPa.

3.3.2 Critical Crack Length

The critical crack length can be calculated using equation 22 below.

$$a_{\mathrm{crit}} = \frac{1}{\pi}\left(\frac{K_{\mathrm{IC}}}{Y\,\sigma_{\mathrm{n}}}\right)^2 \tag{22}$$

Where σ_n is the normal stress on the crack (σ_n = 339.52 MPa) as calculated previously, Y is the geometry factor equal to 1.12 for an edge crack (which a corner weld can be idealised as [1]), K_{IC} is fracture toughness (60 $\mathrm{MPa\,m^{1/2}}$ (Appendix A.6)), and a_{crit} is the critical crack length calculated as 7.92 mm.

3.3.3 Summary of Brittle Fracture

A mode I crack is most likely to form at the bottom of the lug near the weld line, where bending produces the highest tensile stresses. The most dangerous additional stress orientation occurs when it aligns with the operational principal stress, giving a combined normal stress of $\sigma_n = 339.52$ MPa at $\theta_n = -14.5°$, normal to the crack plane. Using the material fracture toughness, the critical crack length was found to be $a_{\mathrm{crit}} = 7.92$ mm, meaning any longer crack is expected to fail unstably in brittle fracture.

3.4 Cyclic Loading

This section assessed the probability of a formed crack growing due to cyclic loading under normal operating conditions. The cyclic loading conditions were from 0 load to full operating conditions and the lug was assessed to discern how many of these loading cycles could be completed before a lug inspection was required.

3.4.1 Detection Probability

Taking a_o = 2 mm, the detection probability of 99% and using equation 23 (from the brief) can be rearranged to give a value for the length of crack detected with the probability mentioned (full rearrangement in Appendix B.8).

$$P_{\mathrm{POD}}(a) = 1 - e^{-\left(\frac{a}{a_0}\right)^4}. \tag{23}$$

Where $P_{\text{POD}}(a)$ is the probability of a crack being detected (99$\%$) and a is the detectable crack length, calculated to be: a_1 = 0.00293 m.

3.4.2 Number of Load Cycles

Paris' equation 24 [1] is used to describe the increase in crack length per load cycle through fatigue crack growth.

$$\frac{\mathrm{d}a}{\mathrm{d}N} = C\,(\Delta K)^m \tag{24}$$

Where the constants C and m (provided in the brief) are 10^{-11} and 4 respectively, and ΔK describes the stress intensity loading factor (Appendix B.9 (assuming small deviatoric and hydrostatic deformations during loading cycles). Paris' equation can be rearranged and integrated (full derivation in Appendix B.10) to find an expression for the number of load cycles as a function of different crack lengths, shown below.

$$N = \frac{10^{11}}{\pi^2\big(Y\,\Delta\sigma\big)^4}\left(\frac{1}{a_1} - \frac{1}{a_2}\right). \tag{25}$$

Where the values for Y = 1.12 (for an edge crack), σ_{op} = 70.05 MPa (principal stress at operating load), and a_1 and a_2 represent the crack lengths that are expected N number of cycles between each other. To calculate the maximum inspection interval, the crack must be large enough to be detected (with 99$\%$ probability) and not have time to grow to half the critical crack length (accounting for a crack safety factor of 2, as specified in the brief). So, $a_1 = 2.93 \times 10^{-3}$ m, as previously calculated, and a_2 = 3.96 mm (accounting for the safety factor of 2). After substituting these values into the equation, N can be found to be 24,000, meaning 24,000 loading cycles can be performed before a crack can grow from 2.93 mm to 3.96 mm and a crack inspection is due.

3.5 Summary of Failure Analysis

The global bending behaviour of the lug resulted in spots of high stress on the top and bottom of the lug, depicted in Figure 6 (Appendix B.11). The top of the lug in compression experienced the highest overall von Mises stress and was analysed here for yield failure under the extra stress conditions described in the brief (σ_p = 269.47 MPa at 17.5°). This highest stress point resulted in an accompanied high strain and therefore justified the placement of a strain gauge here, oriented to align with maximum principal strain (17.5°) to detect unsafe deformation during operation (any compressive strain value greater than $\varepsilon_{crit} = 1688\mu\varepsilon$). Loading the tensile stress hotspot at the bottom of the weld line past yield would result in permanent plastic deformation and eventually a Mode I crack (at 90° to the principal stress). To ensure cyclic loading does not lead to the critical crack length (7.92 mm obtained through brittle fracture analysis), lug inspection should be performed every 24,000 load cycles, via Paris' law (accounting for a safety factor of 2).

References

[1] N. E. Dowling, *Mechanical Behavior of Materials: Engineering Methods for Deformation, Fracture, and Fatigue*, 5th ed., Pearson, Harlow, 2019.

[2] A. Velichko, "Finite Element Analysis – Lecture Notes," MENG30011, University of Bristol, 2025.

[3] Dassault Systèmes, *Abaqus 2024 Documentation*, Version 2024, Dassault Systèmes Simulia Corp., Providence, RI, 2024.

[4] British Standards Institution, *BS EN 1090-2:2018. Execution of steel structures and aluminium structures – Technical requirements for steel structures*, BSI, London, 2018.

[5] HBM, "Y Series Strain Gauges" datasheet, Hottinger Brüel & Kjær, Darmstadt, Germany, 2023. Available: `https://usermanual.wiki/CHM/StrainGaugesdatasheetHBM.2224230565.pdf`

A Task 1 Appendix

A.1 Lug dimensions

$$a = 27 \text{ mm},$$
$$b = 84 \text{ mm},$$
$$c = 34 \text{ mm}.$$

A.2 Shape functions for the 6-node triangular element [2]

Using natural coordinates (s, t),

$$N_1(s,t) = 1 - 3s - 3t + 2s^2 + 2t^2 + 4st, \tag{26}$$
$$N_2(s,t) = -s + 2s^2, \tag{27}$$
$$N_3(s,t) = -t + 2t^2, \tag{28}$$
$$N_4(s,t) = 4s - 4s^2 - 4st, \tag{29}$$
$$N_5(s,t) = 4st, \tag{30}$$
$$N_6(s,t) = 4t - 4t^2 - 4st. \tag{31}$$

A.3 Strain-displacement matrix $\mathrm{B}(s,t)$ [2]

The 3×12 strain-displacement matrix for plane stress is

$$\mathbf{B}(s,t) = \begin{bmatrix} N_{1,x} & 0 & N_{2,x} & 0 & N_{3,x} & 0 & N_{4,x} & 0 & N_{5,x} & 0 \\ N_{6,x} & 0 & & & & & & & & \\ 0 & N_{1,y} & 0 & N_{2,y} & 0 & N_{3,y} & 0 & N_{4,y} & 0 & N_{5,y} \\ 0 & N_{6,y} & & & & & & & & \\ N_{1,y} & N_{1,x} & N_{2,y} & N_{2,x} & N_{3,y} & N_{3,x} & N_{4,y} & N_{4,x} & N_{5,y} & N_{5,x} \\ N_{6,y} & N_{6,x} & & & & & & & & \end{bmatrix}. \tag{32}$$

A.4 Jacobian and inverse Jacobian matrices

The Jacobian matrix $\mathbf{J}$ for the mapping from natural (s, t) to physical (x, y) coordinates is

$$\mathbf{J}(s,t) = \begin{bmatrix} \dfrac{\partial x}{\partial s} & \dfrac{\partial x}{\partial t} \\ \dfrac{\partial y}{\partial s} & \dfrac{\partial y}{\partial t} \end{bmatrix}, \tag{33}$$

and its inverse is

$$\mathbf{J}^{-1}(s,t) = \begin{bmatrix} \dfrac{\partial s}{\partial x} & \dfrac{\partial s}{\partial y} \\ \dfrac{\partial t}{\partial x} & \dfrac{\partial t}{\partial y} \end{bmatrix}. \tag{34}$$

A.5 $x(s,t)$ and $y(s,t)$ mappings

The isoparametric mapping is

$$x(s,t) = \sum_{i=1}^{6} N_i(s,t)\, x_i, \tag{35}$$
$$y(s,t) = \sum_{i=1}^{6} N_i(s,t)\, y_i. \tag{36}$$

A.6 Material properties of S355JR

As provided in the brief

Property	Value
Young's modulus E	209 GPa
Poisson's ratio ν	0.3
Yield strength	355 MPa
Ultimate tensile strength	630 MPa
Elongation at failure	22%
Density	8010 kg m^{-3}
Fracture toughness	60 MPa m$^{1/2}$

A.7 Numerical D matrix

For $E = 209$ GPa and $\nu = 0.3$, the plane stress $\mathbf{D}$ matrix becomes

$$\mathbf{D} \approx \begin{bmatrix} 2.30 \times 10^5 & 6.89 \times 10^4 & 0 \\ 6.89 \times 10^4 & 2.30 \times 10^5 & 0 \\ 0 & 0 & 8.04 \times 10^4 \end{bmatrix} \text{ N/mm}^2. \tag{37}$$

A.8 Full stiffness matrix K

The element stiffness matrix can be written as

$$\mathbf{K} = \frac{Et}{1-\nu^2}\,\mathbf{K}_0, \tag{38}$$

where $t = c$ is the thickness and $\mathbf{K}_0$ is the dimensionless 12×12 matrix

$$\mathbf{K}_0 = \begin{bmatrix} 1.69 & -0.506 & 0.593 & -0.219 & -0.0292 & 0.0500 & -2.37 & 0.874 & 0 & 0 \\ 0.117 & -0.200 & & & & & & & & \\ -0.506 & 0.933 & -0.227 & 0.394 & 0.0583 & -0.0833 & 0.907 & -1.58 & 0 & 0 \\ -0.233 & 0.333 & & & & & & & & \\ 0.593 & -0.227 & 1.92 & -0.831 & 0.0479 & -0.0500 & -2.37 & 0.907 & -0.192 & 0.200 \\ 0 & 0 & & & & & & & & \\ -0.219 & 0.394 & -0.831 & 1.59 & -0.0583 & 0.137 & 0.874 & -1.58 & 0.233 & -0.548 \\ 0 & 0 & & & & & & & & \\ -0.0292 & 0.0583 & 0.0479 & -0.0583 & 0.0563 & 0 & 0 & 0 & -0.192 & 0.233 \\ 0.117 & -0.233 & & & & & & & & \\ 0.0500 & -0.0833 & -0.0500 & 0.137 & 0 & 0.161 & 0 & 0 & 0.200 & -0.548 \\ -0.200 & 0.333 & & & & & & & & \\ -2.37 & 0.907 & -2.37 & 0.874 & 0 & 0 & 4.89 & -1.78 & 0.233 & -0.433 \\ -0.383 & 0.433 & & & & & & & & \\ 0.874 & -1.58 & 0.907 & -1.58 & 0 & 0 & -1.78 & 3.58 & -0.433 & 0.667 \\ 0.433 & -1.10 & & & & & & & & \\ 0 & 0 & -0.192 & 0.233 & -0.192 & 0.200 & 0.233 & -0.433 & 4.89 & -1.78 \\ -4.74 & 1.78 & & & & & & & & \\ 0 & 0 & 0.200 & -0.548 & 0.233 & -0.548 & -0.433 & 0.667 & -1.78 & 3.58 \\ 1.78 & -3.16 & & & & & & & & \\ 0.117 & -0.233 & 0 & 0 & 0.117 & -0.200 & -0.383 & 0.433 & -4.74 & 1.78 \\ 4.89 & -1.78 & & & & & & & & \\ -0.200 & 0.333 & 0 & 0 & -0.233 & 0.333 & 0.433 & -1.10 & 1.78 & -3.16 \\ -1.78 & 3.58 & & & & & & & & \end{bmatrix}. \tag{39}$$

A.9 Reduced stiffness matrix

The reduced stiffness matrix for the free degrees of freedom is

$$\mathbf{K}_{\text{red}} \approx \begin{bmatrix} 1.50\times10^{7} & -6.49\times10^{6} & -1.85\times10^{7} & 7.09\times10^{6} & -1.50\times10^{6} & 1.56\times10^{6} \\ -6.49\times10^{6} & 1.24\times10^{7} & 6.83\times10^{6} & -1.23\times10^{7} & 1.82\times10^{6} & -4.28\times10^{6} \\ -1.85\times10^{7} & 6.83\times10^{6} & 3.82\times10^{7} & -1.39\times10^{7} & 1.82\times10^{6} & -3.38\times10^{6} \\ 7.09\times10^{6} & -1.23\times10^{7} & -1.39\times10^{7} & 2.80\times10^{7} & -3.38\times10^{6} & 5.21\times10^{6} \\ -1.50\times10^{6} & 1.82\times10^{6} & 1.82\times10^{6} & -3.38\times10^{6} & 3.82\times10^{7} & -1.39\times10^{7} \\ 1.56\times10^{6} & -4.28\times10^{6} & -3.38\times10^{6} & 5.21\times10^{6} & -1.39\times10^{7} & 2.80\times10^{7} \end{bmatrix} \text{ N/mm.} \tag{40}$$

A.10 Nodal displacement vector

$$\mathbf{U} = \begin{bmatrix} u_{2x} & u_{2y} & u_{4x} & u_{4y} & u_{5x} & u_{5y} \end{bmatrix}^{\mathrm{T}} \approx \begin{bmatrix} 5.63\times10^{-6} \\ 1.11\times10^{-5} \\ 2.47\times10^{-6} \\ 4.54\times10^{-6} \\ 3.42\times10^{-7} \\ 1.01\times10^{-6} \end{bmatrix}. \tag{41}$$

A.11 Strains at Gauss quadrature points

Using equation (4) at the Gauss points:

Gauss Point 1 $(0.5, 0)$**:**

$$\begin{aligned} \varepsilon_{xx} &\approx 2.09\times10^{-4}, \\ \varepsilon_{yy} &\approx -4.22\times10^{-4}, \\ \gamma_{xy} &\approx 1.89\times10^{-4}. \end{aligned}$$

Gauss Point 2 $(0, 0.5)$**:**

$$\begin{aligned} \varepsilon_{xx} &\approx 0, \\ \varepsilon_{yy} &\approx 0, \\ \gamma_{xy} &\approx 0. \end{aligned}$$

Gauss Point 3 $(0.5, 0.5)$**:**

$$\begin{aligned} \varepsilon_{xx} &\approx 5.06\times10^{-5}, \\ \varepsilon_{yy} &\approx -2.07\times10^{-4}, \\ \gamma_{xy} &\approx 5.66\times10^{-5}. \end{aligned}$$

A.12 Stresses and von Mises stress at Gauss points

Using equations (14) and (15):

Gauss Point 1 $(0.5, 0)$**:**

$$\begin{aligned} \sigma_{xx} &\approx 18.8 \text{ MPa}, \\ \sigma_{yy} &\approx -82.5 \text{ MPa}, \\ \tau_{xy} &\approx 15.2 \text{ MPa}, \\ \sigma_{\mathrm{vm}} &\approx 97.0 \text{ MPa}. \end{aligned}$$

Gauss Point 2 $(0, 0.5)$**:**

$$\begin{aligned}\sigma_{xx} &\approx 0,\\ \sigma_{yy} &\approx 0,\\ \tau_{xy} &\approx 0,\\ \sigma_{\mathrm{vm}} &\approx 0.\end{aligned}$$

Gauss Point 3 $(0.5, 0.5)$**:**

$$\begin{aligned}\sigma_{xx} &\approx -2.61\ \text{MPa},\\ \sigma_{yy} &\approx -44.0\ \text{MPa},\\ \tau_{xy} &\approx 4.55\ \text{MPa},\\ \sigma_{\mathrm{vm}} &\approx 43.4\ \text{MPa}.\end{aligned}$$

A.13 Energy Balance Equations

Strain Energy:

$$U = \frac{1}{2}\,\mathbf{u}_{\text{free}}^{\mathrm{T}}\,\mathbf{K}_{\text{red}}\,\mathbf{u}_{\text{free}} \tag{42}$$

Work Done:

$$W = \frac{1}{2}\,F_{2y}\,u_{2y} \tag{43}$$

Equating both:

$$W = U \tag{44}$$

A.14 Second Moment of Inertia

$$I = \frac{bh^3}{36} \tag{45}$$

B Task 3 Appendix

B.1 Plane-stress transformation matrix [1]

$$\sigma_{\mathsf{p}} = \begin{bmatrix} \sigma_p \cos^2\theta_p & \sigma_p \sin\theta_p \cos\theta_p \\ \sigma_p \sin\theta_p \cos\theta_p & \sigma_p \sin^2\theta_p \end{bmatrix}. \tag{46}$$

B.2 Principal stress equations

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \;\pm\; \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \tag{47}$$

B.3 Principal strain equations

$$\varepsilon_{1,2} = \frac{\varepsilon_x + \varepsilon_y}{2} \;\pm\; \sqrt{\left(\frac{\varepsilon_x - \varepsilon_y}{2}\right)^2 + \varepsilon_{xy}^2} \tag{48}$$

B.4 Angle of principal stress/strain [1]

$$\tan 2\theta_\sigma = \frac{2\tau_{xy}}{\sigma_x - \sigma_y} \tag{49}$$

Taking γ_{xy} as engineering shear strain,

$$\tan 2\theta_\varepsilon = \frac{\gamma_{xy}}{\varepsilon_x - \varepsilon_y} \tag{50}$$

B.5 Von Mises in plane-stress

$$\sigma_{\text{vm}} = \sqrt{\frac{1}{2}\left[(\sigma_1 - \sigma_2)^2 + \sigma_1^2 + \sigma_2^2\right]} \tag{51}$$

B.6 Principal stress-strain relations

Strain to stress:

$$\sigma_1 = \frac{E}{1-\nu^2}\left(\varepsilon_1 + \nu\,\varepsilon_2\right) \tag{52}$$

$$\sigma_2 = \frac{E}{1-\nu^2}\left(\varepsilon_2 + \nu\,\varepsilon_1\right) \tag{53}$$

Stress to strain:

$$\varepsilon_1 = \frac{1}{E}\left(\sigma_1 - \nu\,\sigma_2\right) \tag{54}$$

$$\varepsilon_2 = \frac{1}{E}\left(\sigma_2 - \nu\,\sigma_1\right) \tag{55}$$

B.7 General Relations

$$\varepsilon_x = \frac{1}{E}\left(\sigma_x - \nu\sigma_y\right) \tag{56}$$

$$\varepsilon_y = \frac{1}{E}\left(\sigma_y - \nu\sigma_x\right) \tag{57}$$

$$\sigma_x = \frac{E}{1-\nu^2}\left(\varepsilon_x + \nu\,\varepsilon_y\right), \tag{58}$$

$$\sigma_y = \frac{E}{1-\nu^2}\left(\varepsilon_y + \nu\,\varepsilon_x\right). \tag{59}$$

$$\gamma_{xy} = \frac{\tau_{xy}}{G} \tag{60}$$

$$G = \frac{E}{2(1+\nu)} \tag{61}$$

B.8 Derivation of a_1

Starting from the probability of detection (POD) expression provided in the brief:

$$P_{\mathrm{POD}}(a) = 1 - \exp\left[-\left(\frac{a}{a_0}\right)^4\right], \tag{62}$$

with $a_0 = 2\ \mathrm{mm}$ and $P_{\mathrm{POD}} = 0.99$.
Substituting these values:

$$0.99 = 1 - \exp\left[-\left(\frac{a}{2}\right)^4\right], \tag{63}$$

Rearranged to isolate the exponential term:

$$0.01 = \exp\left[-\left(\frac{a}{2}\right)^4\right], \tag{64}$$

Taking natural logarithms of both sides:

$$\ln(0.01) = -\left(\frac{a}{2}\right)^4, \tag{65}$$

Multiplying by -1:

$$-\ln(0.01) = \left(\frac{a}{2}\right)^4, \tag{66}$$

Taking the fourth root:

$$\frac{a}{2} = \left[-\ln(0.01)\right]^{1/4}, \tag{67}$$

Solving for a:

$$a = 2\ \left[-\ln(0.01)\right]^{1/4}. \tag{68}$$

Evaluating this expression numerically:

$$a_1 = 2.93\ \mathrm{mm} = 0.00293\ \mathrm{m}. \tag{69}$$

B.9 Stress Intensity Factor

Assuming small deviatoric and hydrostatic deformations during loading cycles.

$$\Delta K = Y\,\Delta\sigma\,\sqrt{\pi a} \tag{70}$$

B.10 Paris' Law Derivation

Paris' law [1]:

$$\frac{\mathrm{d}a}{\mathrm{d}N} = C\,(\Delta K)^m, \tag{71}$$

where

$$C = 10^{-11}, \qquad m = 4. \tag{72}$$

For mode I loading, the stress intensity factor range is:

$$\Delta K = Y\,\Delta\sigma\,\sqrt{\pi a}. \tag{73}$$

Substituting ΔK into Paris' law:

$$\frac{\mathrm{d}a}{\mathrm{d}N} = 10^{-11} \left(Y\,\Delta\sigma\,\sqrt{\pi a}\right)^4 \tag{74}$$

$$= 10^{-11} \left(Y\,\Delta\sigma\right)^4 \left(\sqrt{\pi a}\right)^4 \tag{75}$$

$$= 10^{-11} \left(Y\,\Delta\sigma\right)^4 \pi^2 a^2. \tag{76}$$

Rearranging for $\mathrm{d}N$:

$$\mathrm{d}N = \frac{\mathrm{d}a}{10^{-11} \left(Y\,\Delta\sigma\right)^4 \pi^2 a^2}. \tag{77}$$

Integrating from a_1 to a_2:

$$N = \int_{a_1}^{a_2} \frac{\mathrm{d}a}{10^{-11} \left(Y\,\Delta\sigma\right)^4 \pi^2 a^2} \tag{78}$$

$$= \frac{1}{10^{-11} \left(Y\,\Delta\sigma\right)^4 \pi^2} \int_{a_1}^{a_2} a^{-2}\,\mathrm{d}a. \tag{79}$$

Evaluating the integral:

$$\int a^{-2}\,\mathrm{d}a = -a^{-1}, \tag{80}$$

so

$$N = \frac{1}{10^{-11} \left(Y\,\Delta\sigma\right)^4 \pi^2} \left[-a^{-1}\right]_{a_1}^{a_2} \tag{81}$$

$$= \frac{1}{10^{-11} \left(Y\,\Delta\sigma\right)^4 \pi^2} \left(-\frac{1}{a_2} + \frac{1}{a_1}\right) \tag{82}$$

$$\tag{83}$$

Written compactly as:

$$N = \frac{10^{11}}{\pi^2 \left(Y\,\Delta\sigma\right)^4} \left(\frac{1}{a_1} - \frac{1}{a_2}\right). \tag{84}$$

B.11 Figure 6 Diagram of Applied Stresses and Relevant Nodes

B.12 Figure 7 Graph of σ_p loaded at angle θ_p required for yield

Where the dotted line is the minimum required tension for yielding to occur

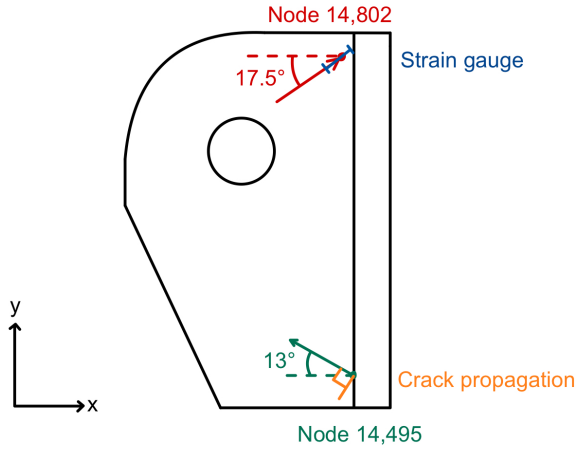


Figure 6: Diagram of Applied Stresses and relevant nodes

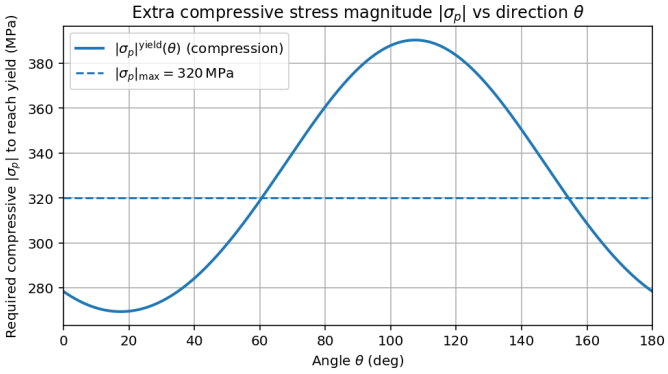


Figure 7: σ_p required for failure at angle θ_p